

DPP - 3

- Q.1 Let A & B be two events. Suppose $P(A) = 0.4$, $P(B) = p$ & $P(A \cup B) = 0.7$. The value of p for which A & B are independent is :
(A) $1/3$ (B) $1/4$ (C) $1/2$ (D) $1/5$
- Q.2 A pair of numbers is picked up randomly (without replacement) from the set $\{1, 2, 3, 5, 7, 11, 12, 13, 17, 19\}$. The probability that the number 11 was picked given that the sum of the numbers was even, is nearly :
(A) 0.1 (B) 0.125 (C) 0.24 (D) 0.18
- Q.3 For a biased die the probabilities for the different faces to turn up are given below :
Faces : 1 2 3 4 5 6
Probabilities : 0.10 0.32 0.21 0.15 0.05 0.17
The die is tossed & you are told that either face one or face two has turned up. Then the probability that it is face one is :
(A) $1/6$ (B) $1/10$ (C) $5/49$ (D) $5/21$
- Q.4 A determinant is chosen at random from the set of all determinants of order 2 with elements 0 or 1 only. The probability that the determinant chosen has the value non negative is :
(A) $3/16$ (B) $6/16$ (C) $10/16$ (D) $13/16$
- Q.5 15 coupons are numbered 1, 2, 3,....., 15 respectively. 7 coupons are selected at random one at a time with replacement. The probability that the largest number appearing on a selected coupon is 9 is :
(A) $\left(\frac{9}{16}\right)^6$ (B) $\left(\frac{8}{15}\right)^7$ (C) $\left(\frac{3}{5}\right)^7$ (D) $\frac{9^7 - 8^7}{15^7}$
- Q.6 A card is drawn & replaced in an ordinary pack of 52 playing cards. Minimum number of times must a card be drawn so that there is atleast an even chance of drawing a heart, is
(A) 2 (B) 3 (C) 4 (D) more than four
- Q.7 A license plate is 3 capital letters (of English alphabets) followed by 3 digits. If all possible license plates are equally likely, the probability that a plate has either a letter palindrome or a digit palindrome (or both), is
(A) $\frac{7}{52}$ (B) $\frac{9}{65}$ (C) $\frac{8}{65}$ (D) none
- Q.8 Whenever horses a, b, c race together, their respective probabilities of winning the race are 0.3, 0.5 and 0.2 respectively. If they race three times the probability that "the same horse wins all the three races" and the probability that a, b, c each wins one race, are respectively (Assume no dead heat)
(A) $\frac{8}{50}$; $\frac{9}{50}$ (B) $\frac{16}{100}$, $\frac{3}{100}$ (C) $\frac{12}{50}$; $\frac{15}{50}$ (D) $\frac{10}{50}$; $\frac{8}{50}$
- Q.9 Two cubes have their faces painted either red or blue. The first cube has five red faces and one blue face. When the two cubes are rolled simultaneously, the probability that the two top faces show the same colour is $1/2$. Number of red faces on the second cube, is
(A) 1 (B) 2 (C) 3 (D) 4

- Q.10 A committee of three persons is to be randomly selected from a group of three men and two women and the chair person will be randomly selected from the committee. The probability that the committee will have exactly two women and one man, and that the chair person will be a woman, is/are
 (A) $1/5$ (B) $8/15$ (C) $2/3$ (D) $3/10$
- Q.11 An urn contains 3 red balls and n white balls.
 Mr. A draws two balls together from the urn. The probability that they have the same colour is $1/2$.
 Mr. B draws one ball from the urn, notes its colour and replaces it. He then draws a second ball from the urn and finds that both balls have the same colour is, $5/8$. The possible value of n is
 (A) 9 (B) 6 (C) 5 (D) 1
- Q.12 The probability that an automobile will be stolen and found within one week is 0.0006. The probability that an automobile will be stolen is 0.0015. The probability that a stolen automobile will be found in one week is
 (A) 0.3 (B) 0.4 (C) 0.5 (D) 0.6

[REASONING TYPE]

- Q.13 In one day test match between India and Australia the umpire continues tossing a fair coin until the two consecutive throws either H T or T T are obtained for the first time. If it is H T, India wins and if it is T T, Australia wins.
Statement-1: Both India and Australia have equal probability of winning the toss.
because
Statement-2: If a coin is tossed twice then the events HT or TT are equiprobable.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false. (D) Statement-1 is false, statement-2 is true.

[SUBJECTIVE]

- Q.14 A certain team wins with probability 0.7, loses with probability 0.2 and ties with probability 0.1. The team plays three games. Find the probability
 (i) that the team wins at least two of the games, but lose none.
 (ii) that the team wins at least one game.
- Q.15 The probability that a person will get an electric contract is $2/5$ and the probability that he will not get plumbing contract is $4/7$. If the probability of getting at least one contract is $2/3$, what is the probability that he will get both?
- Q.16 Five horses compete in a race. John picks two horses at random and bets on them. Find the probability that John picked the winner. Assume no dead heat.
- Q.17 There are 6 red balls and 6 green balls in a bag. Five balls are drawn out at random and placed in a red box. The remaining seven balls are put in a green box. If the probability that the number of red balls in the green box plus the number of green balls in the red box is not a prime number, is $\frac{p}{q}$ where p and q are relatively prime, then find the value of $(p + q)$
- Q.18 The odds that a book will be favourably reviewed by three independent critics are 5 to 2, 4 to 3, and 3 to 4 respectively. What is the probability that of the three reviews a majority will be favourable?
- Q.19 When three cards are drawn from a standard 52-card deck, what is the probability they are all of the same rank? (e.g. all three are kings).
- Q.20 A and B in order draw alternatively from a purse containing 3 rupees and 4 nP's, find their respective chances of first drawing a rupee, the coins once drawn not being replaced.